FOURIRER TRANSFORM

→ FOR APERIODIC SIGNAL

FOURIER PAIR

- **FOURIER TRANSFORM** $S(f) = \int_{-\infty}^{+\infty} s(t) e^{-j2\pi f t} dt$ ← FROM TIME DOMAIN TO FREQUENCY DOMAIN
- **FOURIER ANTITRANSFORM** $s(t) = \int_{-\infty}^{+\infty} S(f) e^{j2\pi f t} df$ ← FROM FREQUENCY DOMAIN TO TIME DOMAIN

APERIODIC SIGNAL

- NO SPECIAL FREQUENCY → ALL FREQUENCY CONTRIBUTE

COSINE MULTIPLICATION

WHEN WE MULTIPLY A SIGNAL $s(t)$ BY A SINUSOIDAL FUNCTION $\cos(\omega_0 t)$ WE GET

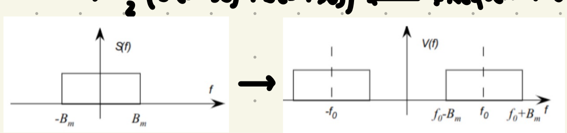
$$v(t) = s(t)\cos(\omega_0 t)$$

$$\cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \longrightarrow$$

$s(t)$ IS BEING MODULATED BY TWO SEPARATE EXPONENTIAL SIGNALS CREATING $v(t)$

WE HAVE A FREQUENCY SHIFT OF THE ORIGINAL SIGNAL

$$V(f) = \frac{1}{2}\left(S(f-f_0) + S(f+f_0)\right)$$ ← FREQUENCY SHIFT OF f_0



CENTRED IN 0 → CENTRED IN $\pm f_0$

⚠ THE SHAPE OF THE SIGNAL DOES NOT CHANGE

- **FREQUENCY DIVISION MULTIPLE ACCESS (FDMA)**
  - MULTIPLE SIGNALS ARE TRANSMITTED AT THE SAME TIME USING DIFFERENT FREQUENCY BANDS FOR EACH SIGNAL
  - COMBINE SIGNAL IN A COMPOSITE SIGNAL FOR TRANSMISSION
  - AT RECEIVING DIFFERENT SIGNAL COMPONENTS ARE SEPARATED BASED ON THEIR FREQUENCY BANDS
- **FREQUENCY DIVISION DUPLEX (FDD)**
  - TRANSMIT DIFFERENT SIGNALS SIMULTANEOUSLY IN BOTH DIRECTIONS (FORWARD - BACKWARD)
  - SIMULTANEOUS TRANSMISSION AND RECEPTION OF SIGNALS EACH IN SEPARATE FREQUENCY BAND
- **FREQUENCY REUSE IN CELLULAR LINES**
  - UNIQUE FREQUENCY SETS IN ADJACENT CELLS, REUSED TO MINIMIZE INTERFERENCE
- **GSM (2G)**
  - SPECIFIC REUSE PATTERNS TO MINIMIZE INTERFERENCE
- **LTE (4G)**
  - DYNAMIC FREQUENCY ALLOCATION FOR IMPROVED PERFORMANCE

PROPERTIES OF FOURIRER TRANSFORM

- **LINEARITY:** THE TRANSFORM OF THE LINEAR COMBINATION OF TWO APERIODIC SIGNALS IS EQUAL TO THE LINEAR COMBINATION OF THE TRANSFORM OF THE ORIGINAL SIGNAL

$$a s_1(t) + b s_2(t) \longrightarrow a S_1(f) + b S_2(f)$$

- **DELAY/ANTICIPATION:** THE TRANSFORM OF THE TIME-SHIFTED SIGNALS IS EQUAL TO THE TRANSFORM OF THE ORIGINAL SIGNAL MULTIPLIED BY A COMPLEX EXPONENTIAL

$$s(t) \longrightarrow S(f) \qquad s(t-t_1) \longrightarrow S(f) \cdot e^{-j2\pi f t_1}$$

- **PHASE SHIFT (FREQ. SHIFT):** MULTIPLY A SIGNAL BY A COMPLEX EXPONENTIAL (WITH FREQUENCY MULTIPLE OF THE FUNDAMENTAL ONE), RESULT IN A FREQUENCY SHIFT OF THE FOURIRER TRANSFORM OF THE NEW SIGNAL

$$s(t) \longrightarrow S(f) \qquad s(t)\, e^{j2\pi f_1 t} \longrightarrow S(f-f_1)$$

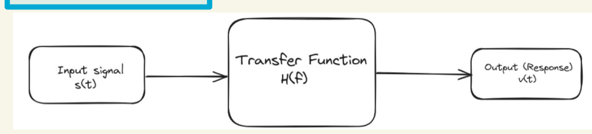
- **INTEGRAL:** THE TRANSFORM OF THE INTEGRAL OVER THE PERIOD OF SIGNAL ARE MULTIPLIED BY A FACTOR OF $-j\frac{1}{2\pi f}$ INVERSELY PROPORTIONAL TO FREQUENCY

$$s(t) \longrightarrow S(f) \longrightarrow \int_{-\infty}^{t} s(t) dt \longrightarrow -j\frac{1}{2\pi f} S(f)$$

- **DERIVATE:** THE TRANSFOR OF THE DERIVATE OVER TIME OF A SIGNAL IS MULTIPLIED BY A COMPLEX COEFFICIENT DIRECTLY PROPORTIONAL TO FREQUENCY

$$s(t) \longrightarrow S(f) \longrightarrow \frac{d}{dt} s(t) \longrightarrow j2\pi f S(f)$$

CONVOLUTION



PASSING A SIGNAL $s(t)$ THROUGH A LINEAR SYSTEM CORRESPONDS TO MULTIPLY ITS TRANSFORM $S(f)$ BY THE BLOCK'S TRANSFER FUNCTION $H(f)$

$$V(f) = S(f) H(f)$$

IN TIME DOMAIN: $v(t) = s(t) \underset{\text{CONVOLUTION}}{*} \overset{\text{ANTITRASFORM of } H(f)}{h(t)}$

CONVOLUTION THEOREM: THE FOURIER TRANSFORM OF A CONVOLUTION OF TWO FUNCTION IS EQUAL TO THE PRODUCT OF THEIR FOURIER TRANSFORM

THE CONVOLUTION BETWEEN A FUNCTION AND DIRAC'S DELTA IS THE FUNCTION ITSELF

PARSEVAL THEOREM

ENERGY OF A SIGNAL IN TIME DOMAIN

$$E = \int_{-\infty}^{+\infty} s^2(t) dt$$

ENERGY OF A SIGNAL IN FREQUENCY DOMAIN

$$E = \int_{-\infty}^{+\infty} S(f) S(f)^* df = \int_{-\infty}^{+\infty} |S(f)|^2 df$$

SPECTRAL ENERGY DENSITY

$$G(f) = |S(f)|^2 \longrightarrow E = \int_{-\infty}^{+\infty} G(f) df$$

IF WE CONSIDER ONLY THE INTERVAL BETWEEN TWO FREQUENCIES, THE ENERGY OF THE SIGNAL BETWEEN THE TWO FREQUENCIES IS

$$E(f_1; f_2) = \int_{f_1}^{f_2} |S(f)|^2 df + \int_{-f_2}^{-f_1} |S(f)|^2 df$$

ENERGY THROUGH A LINEAR BLOCK

- **FREQUENCY DOMAIN**: IF AN INPUT PASSES THROUGH A LINEAR SYSTEM (BLOCK) $\longrightarrow Y(f) = H(f)\,S(f)$
- **TIME DOMAIN**: IF AN INPUT PASSES THROUGH A LINEAR SYSTEM (BLOCK) $\longrightarrow y(t) = s(t) * h(t)$

SPECTRAL DENSITY AT THE OUTPUT $\longrightarrow G_o(f) = |H(f)|^2\, G(f)$

DELAY BLOCK $h(t) = \delta(t - T_d) \longrightarrow H(f) = e^{-j 2\pi f T_d}$

- A DELAY BLOCK SHIFT THE SIGNAL IN TIME, BUT DOES NOT ALTER THE AMPLITUDE. IT PRESERVE THE SIGNAL ENERGY $(E_o = E)$ AND IT AFFECT ONLY THE PHASE

Energy Through a Linear Block

- In the frequency domain, if an input passes through a linear system (block), we have a product ($Y(f) = H(f)S(f)$)
- In the TIME domain, if an input passes through a linear system (block), we have a convolution operation ($y(t) = s(t) * h(t)$)

The spectral density of energy at the output $G_o(f)$ is related to the spectral density at the input $G(f)$:

$$G_o(f) = |S_o(f)|^2 = |H(f)|^2 |S(f)|^2 = |H(f)|^2 G(f)$$

-> then, a delay block does not change the energy, because it operates only on the phase:

$$h(t) = \delta(t - T_D) \quad \Rightarrow \quad H(f) = e^{-j2\pi f T_D} \quad \Rightarrow \quad E_0 = E$$

-> In contrast, a block that does not distort (=does not change) the shape of the output signal can change its energy (e.g., amplifier):

$$h(t) = h_0 \delta(t) \quad \Rightarrow \quad H(f) = h_0 \quad \Rightarrow \quad s(t) = h_0 s(t) \quad \Rightarrow \quad E_0 = h_0^2 E$$